



DOI: 10.1111/1471-0528.16524
www.bjog.org

Systematic Review

Duration of estrogen exposure during reproductive years, age at menarche and age at menopause, and risk of cardiovascular disease events, all-cause and cardiovascular mortality: a systematic review and meta-analysis

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Accepted 25 August 2020. Published Online 5 November 2020.

Background Little is known about the estrogen exposure measurement and mutual effect of age at menarche and age at menopause in the risk of cardiovascular disease (CVD) events.

Objectives To evaluate estrogen exposure measurement and describe mutual effect of age at menarche and age at menopause in the risk of CVD events.

Search strategy Systematic review of literature in PubMed, Embase and Web of Science for studies published up to 28 June 2020.

Selection criteria Observational studies related to estrogen exposure measurement, including mutual effect of age at menarche and age at menopause and risk of CVD events.

Data collection and analysis Synthesis of evidence was conducted by reviewing individual estimates, followed by meta-analysis. The study received no external funding.

Main results A total of 75 studies were included in synthesis of evidence, of which 17 studies were included in meta-analysis. Reproductive lifespan (age at menopause – age at

menarche), endogenous estrogen exposure and total estrogen exposure were used for estrogen exposure measurement. Reproductive lifespan was by far the most commonly used method for estrogen exposure measurement. A shorter reproductive lifespan was associated with a higher risk of CVD events; the pooled relative risk (95% CI) was 1.31 (1.25–1.36) for stroke events. Robust epidemiological studies with measurement of estrogen exposure and associated health risk would strengthen the evidence.

Conclusions Reproductive lifespan was the most commonly used method for estrogen exposure measurement in epidemiological studies. A shorter reproductive lifespan was associated with a higher risk of CVD events, particularly stroke.

Keywords Cardiovascular diseases, estrogen exposure, menarche, menopause, mortality.

Tweetable abstract A systematic review and meta-analysis found that women with a shorter reproductive lifespan have a higher risk of stroke events.

Please cite this paper as: Mishra SR, Chung H-F, Waller M, Mishra GD. Duration of estrogen exposure during reproductive years, age at menarche and age at menopause, and risk of cardiovascular disease events, all-cause and cardiovascular mortality: a systematic review and meta-analysis. BJOG. 2021;128:809–821.

Introduction

Cardiovascular diseases (CVD) are the leading cause of deaths and disability among women worldwide.¹ From 2006 to 2016, the burden of CVDs increased by 14.4% among women.¹ Published studies have shown an association

between the timing of menarche and menopause and CVD events, all-cause and cardiovascular mortality.^{2,3} However, there was insufficient evidence to draw conclusions on the role of reproductive lifespan on these health outcomes.

Reproductive lifespan can be described as the difference between age at menarche and age at menopause, which is a simple way to estimate estrogen exposure.^{4–7} Some studies further take into account the relevant reproductive

PROSPERO registration: CRD42019126655

processes such as pregnancy (including breastfeeding, stillbirths and miscarriages) and hormone use (oral contraceptive pills [OCP] and menopausal hormone therapy [MHT]) to estimate endogenous and total estrogen exposure.^{8,9} For instance, it is known that levels of endogenous estrogen rise during pregnancy, but fall after birth, particularly with breastfeeding. An earlier study found that parous women with four or more children had 20% less concentration of free estradiol compared with women with one child, but the level of estrone, another derivative of endogenous estrogen, did not differ with parity.¹⁰ Breastfeeding,¹¹ miscarriages, stillbirths and induced termination of pregnancy¹² causes changes in estrogen levels in the human body and their effect is likely to extend beyond CVD.⁸

If the studies do not calculate the total duration between menarche and menopause, an alternative method is to adjust for the timing of both menarche and menopause in the same analysis, which allows benchmarking these two key reproductive events. These indirect measures of estrogen exposure are important for population-level study because the direct measures of hormone exposure are difficult to ascertain. Further, our recent study conducted among 51 450 postmenopausal women from the InterLACE consortium found that timing of menarche and menopause are inter-related: women who experienced early menarche were at higher risk of premature menopause (relative risk ratio 1.80, 95% CI 1.53–2.12) and early menopause (1.31, 95% CI 1.19–1.44).¹³ Therefore, we examined the estrogen exposure measurements and described mutual effect of age at menarche and age at menopause on the risk of CVD events and mortality accompanied by a meta-analysis.

Methods

A systematic review was conducted following the Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. Two independent reviewers (SRM, HFC) conducted the review of literature in PubMed (from inception), Embase (from 1974) and Web of Science (from inception) for studies published up to 28 June 2020, and evaluated the quality of the studies. We used a set of search terms related to estrogen exposure, menarche and menopause, which were combined with CVD events, all-cause mortality and CVD mortality (see Supplementary material, Appendix S1). Search strategy, study selection, statistical analysis and quality evaluation are described in detail in the Supplementary material (Appendix S2 and Table S1). Patient involvement and core outcome set are not relevant to this study, as we used a systematic review and meta-analysis of published studies. The study received no external funding.

Results

Identification of relevant studies

Our search strategy identified 3908 relevant articles through searches from PubMed/Medline, Embase and Web of Science (Figure 1). After the initial screening, 230 articles were included for full text review, of which 75 studies met the inclusion criteria: 26 studies examining estrogen exposure, 34 studies examining age at menarche and 60 studies examining age at menopause.

Definitions of estrogen exposure

Estrogen exposure

Key characteristics of the studies are summarised in the Supplementary material (Table S2). Nearly a third of studies were from Europe and two-thirds were prospective in design. Three different definitions of estrogen exposure were identified: reproductive lifespan (RLS), endogenous estrogen exposure (EEE) and total estrogen exposure (TEE).

A total of 22 studies calculated the interval between age at menarche and age at menopause, defined as RLS. Further, three studies calculated the RLS and further subtracted the time spent in pregnancy, breastfeeding, stillbirth/miscarriage and OCP use.^{9,14,15} Pregnancy is a period marked by sustained rise in both endogenous estrogen and progesterone whereas breastfeeding and stillbirth/miscarriages are the periods with low endogenous estrogen. The oral contraceptive pills before 1975 had a very high dose of estrogen and progesterone that suppresses the normal cycle of endogenous estrogen.⁹ Therefore, the calculation of endogenous estrogen exposure is normally limited to duration, which is unaffected by the opposing effect of progesterone. Those studies justified the subtraction of the postovulatory period because the period is marked by a rise in progesterone level whereas level of estrogen is relatively low.⁹

The total estrogen exposure considers both endogenous estrogen exposure as well as hormone replacement therapy. We found two algorithms to calculate TEE, one that adds OCP and pregnancy (TEE1: Age at menopause – age at menarche + OCP + pregnancy + HRT), another that subtracts OCP and pregnancy [TEE2: Age at menopause – age at menarche – OCP – pregnancy + HRT].¹⁶ By inclusion of OCP years and duration of MHT, TEE2 is longer compared with EEE and TEE1.

Menarche adjusting for menopausal age/status

Of 34 studies examining age at menarche, only a third of studies (12 studies) adjusted for age at menopause/status. We only described the results for those adjusting for menopausal age/status (see Supplementary material, Table S3). A summary of the results for studies without adjusting for

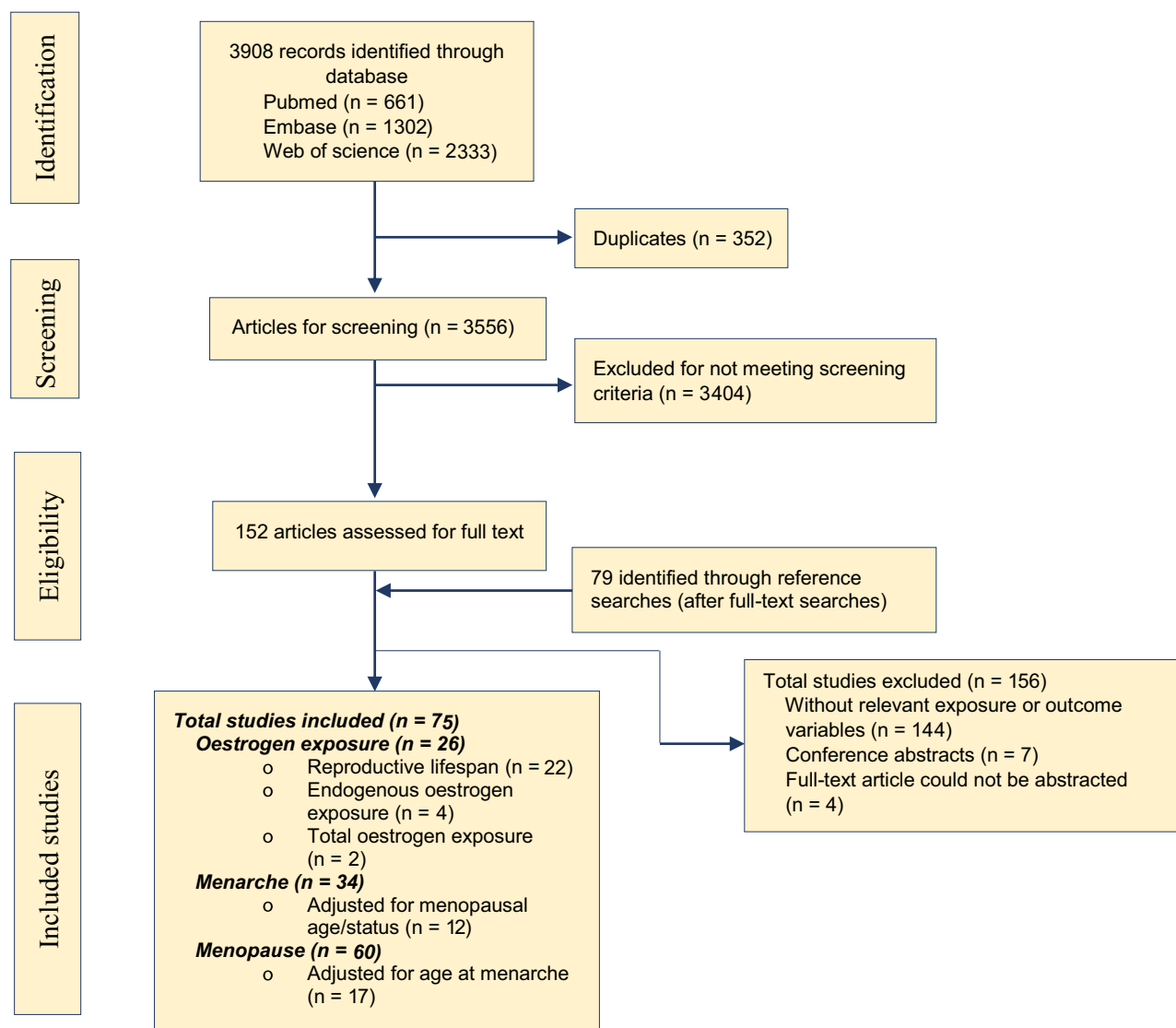


Figure 1. Flow chart of study process.

menopausal age/status is available in the Supplementary material (Table S3).

Menopause adjusting for age at menarche

We reviewed 60 studies that examined age at menopause/status, with only a quarter of studies (17 studies) adjusted for age at menarche. We only described the results for studies adjusted for age at menarche, and a summary of the results for studies without adjusting for age at menarche is available in Table S4 (see Supplementary material). Age at menopause was defined as age at last menstruation for natural menopause in nine studies, age at hysterectomy/oophorectomy for surgical menopause in three studies, whereas for the rest of the studies it was either natural or surgical menopause or undefined.

Association of reproductive lifespan, menarche and menopause with CVD events

Reproductive lifespan

Studies showed association between RLS and overall CVD,^{6,7} coronary heart disease (CHD)^{5,6,17} and stroke events.^{4–7} Findings from these studies are summarised in Table 1.

A higher risk of overall CVD events among women who had shorter RLS was reported by the Nurses' Health Study of 83 169 participants and one cross-sectional study of 3081 participants from the USA.^{6,7} The Nurses' Health Study found that the risk of CVD event was 25% higher among those with <30 years of RLS compared with 34–37 years.⁶ The findings from this and other studies were

Table 1. Summary of studies examining associations between reproductive lifespan, age at menarche and age at menopause with cardiovascular disease events

Outcomes	Short reproductive lifespan	Long reproductive lifespan	Early menarche	Late menarche	Early menopause	Late menopause
CVD (<i>n</i> = 3 studies)						
Ley (2017) ⁶	↑ (<30 vs 34–37 years)	↓ (≥42 vs 34–37 years)*				
Mansoor (2017) ⁷	↑ (≤30 vs 36–40 years)	↔ (>45 vs 36–40 years)			↔ (≤46 vs 50 years)	↔ (per year) ↓ (≥53 vs 50 years)*
Qiu (2013) ²⁰			↔ (9–14 vs 16 years)	↓ (≥18 vs 16 years)		
CHD (<i>n</i> = 5 studies)						
Jung (2016) ^{a5}	↑ (<30 vs 36–39 years)	↑ (≥40 vs 36–39 years)	↔ (≤12 vs 13–14 years)	↔ (≥17 vs 13–14 years)	↑ (≤45 vs 49–51 years)*	↓ (≥52 vs 49–51 years)*
Kim (2016) ⁵¹		↔ (mean diff.)				
Ley (2017) ⁶	↑ (<30 vs 34–37 years)	↓ (≥42 vs 34–37 years)*				
Yang (2017) ¹⁷		↓ (per year)			↑ (<40 vs ≥ 55 years)	↓ (per year) ↓ (per year)
Dam (2019) ²¹						
Myocardial infarction (<i>n</i> = 5 studies)						
Bertuccio (2007) ⁵²			↔ (≤12 vs 13 years)	↔ (≥15 vs 13 years)		
Jung (2016) ⁵	↑ (<30 vs 36–39 years)	↔ (≥40 vs 36–39 years)	↔ (≤12 vs 13–14 years)	↑ (≥17 vs 13–14 years)	↔ (≤45 vs 49–51 years)	↔ (≥52 vs 49–51 years)
Mansoor (2017) ⁷		↔ (>30 vs ≤30 years)				
Palmer (1992) ^{b22}			↔ (≤11 vs 12–14 years)	↔ (≥15 vs 12–14 years)	↑ (<45 vs ≥50 years)	
Saltiki (2006) ⁴¹		↓ (per year)				
Angina (<i>n</i> = 2 studies)						
Mansoor (2017) ⁷		↓ (>30 vs ≤30 years)				
Saltiki (2006) ⁴¹		↔ (per year)				
Heart failure (<i>n</i> = 3 studies)						
Applah (2016) ²³					↑ (<45 vs 50–54 years)	↓ (≥55 vs 50–54 years)*
Halls (2017) ⁵³		↓ (per year)				
Rahman (2015) ²⁴					↑ (40–45 vs 50–54 years)	↔ (≥55 vs 50–54 years)
Stroke (<i>n</i> = 8 studies)						
Alonso (2007) ⁴	↔ (<34 vs 34–36 years)	↓ (>40 vs 34–36 years)	↑ (<12 vs 13 years)	↔ (≥14 vs 13 years)	↔ (>47 vs 51–53 years)	↔ (>53 vs 51–53 years)
Hsieh (2010) ^{c18}	↔ (<32 vs 33–36 years)	↔ (≥37 vs 33–36 years)				
Jung (2016) ^{d5}	↑ (<30 vs 36–39 years)*	↓ (≥40 vs 36–39 years)*	↓ (≤12 vs 13–14 years)*	↑ (≥17 vs 13–14 years)	↑ (≤45 vs 49–51 years)*	↓ (≥52 vs 49–51 years)*
Ley (2017) ⁶	↑ (<30 vs 34–37 years)	↔ (≥42 vs 34–37 years)	↑ (≤13 vs 15 years)	↔ (≥16 vs 15 years)	↑ (46–49 vs 50 years)	↔ (≥51 vs 50 years)
Murakami (2016) ¹⁹		↓ (>30 vs ≤30 years)				
Mansoor (2017) ⁷		↔ (per year)				
Yang (2017) ¹⁷						↓ (per year)

Table 1. (Continued)

Outcomes	Short reproductive lifespan	Long reproductive lifespan	Early menarche	Late menarche	Early menopause	Late menopause
Yang (2019) ⁶⁴²	↑(<30 vs 35–39 years)*	↑(>40 vs 35–39 years)*				
Other CVDs (n = 6 studies)						
Arterial fibrillation						
Wong (2017) ⁵⁴					↔ (<45 vs 45–49 years)	↔ (>54 vs 45–49 years)
Yang (2019) ⁴²	↑(<30 vs ≥ 30 years)				↑(<45 vs ≥ 45 years)	
Carotid plaques						
Stockl (2014) ⁵⁵		↔ (>35 vs ≤35 years)				
Peripheral artery disease						
Corte (2018) ⁵⁶		↔ (per year)				
Holzer (2007) ⁴³		↓ (per year)				
Stockl (2013) ⁴⁴	↔ (<34 vs >39 years)					

↑Indicates the risk of outcome increased, ↓ indicates the risk of outcome decreased, ↔ indicates no evidence of association based on statistical significance at $P < 0.05$.

a: ischaemic heart disease, b: natural menopause, c: ischaemic stroke, d: all stroke, e: among women without arterial fibrillation.

*Trend of ordered exposure levels based on statistical significance at $P < 0.05$.

consistent for CHD.^{5,6,17} However, Jung et al. found a U-shaped association between RLS and ischaemic heart disease; the risk of which increased in both RLS <30 years (40% higher) and RLS ≥40 years (39% higher) (referent: RLS 36–39 years).⁵ Mansoor et al. showed that each year increase in RLS was associated with a 3% reduction in the risk for overall CVD and stroke events.⁷ No effect modification was found by body mass index (BMI) ($P_{\text{interac}} = 0.14$) and smoking variable ($P_{\text{interaction}} = 0.44$).⁷ Except for thrombotic stroke ($P_{\text{trend}} = 0.0004$)⁵ there was no clear evidence of association for other stroke types: haemorrhagic ($P_{\text{trend}} = 0.2197$)⁵ and ischaemic stroke ($P_{\text{trend}} > 0.05$).¹⁸ For other CVD events, such as myocardial infarction, angina, heart failure and peripheral artery disease, there was not sufficient evidence for an association with reproductive lifespan (Table 1).

Age at menarche adjusting for age at menopause

Overall, two studies found that earlier menarche was associated with higher risk of stroke after adjustment for age at menopause.^{4,19} Of these studies, Alonso de Lecinana et al. found that the odds of stroke were 54% higher among women aged <12 years at menarche compared with those aged 13 years at menarche.⁴ Late menarche was not associated with stroke in both studies^{4,19} (Table 1).

For CVD events, a study from China found a lower risk with increasing ages at menarche ($P_{\text{trend}} = 0.049$).²⁰ The risk was 29% lower among those aged >18 years compared with those aged 16 years at menarche. Adjusting for BMI and waist circumference did not attenuate the association (odds ratio 0.71, 95% CI 0.57–0.88 to odds ratio 0.71, 95% CI 0.57–0.89).²⁰

Age at menopause adjusting for age at menarche

Overall, there was some evidence to suggest an association between early menopause and a higher risk of CVD events^{5,21–24} and between late menopause and a lower risk of CVD events^{5,17,20,21,23} after adjustment for age at menarche (Table 1).

A large study comprising 66 104 Korean women found an increasing risk of atherosclerotic CVD with decreasing age at menopause ($P_{\text{trend}} = 0.0176$).⁵ The risk was 18% higher in women aged 46–48 years at menopause compared with those aged 49–51 years and the risk was 11% higher in those aged ≤45 years compared with those aged 49–51 years at menopause.⁵ The association for ischaemic heart disease followed a similar trend: the risk was 28% higher among women aged ≤ 45 compared with those aged 49–51 years ($P_{\text{trend}} = 0.0217$).⁵ Furthermore, two studies showed that early menopause was associated with a higher risk of heart failure.^{23,24}

For stroke, studies showed that early menopause was associated with a higher risk of stroke,^{5,19} and similarly late

menopause was associated with a reduced risk of stroke.^{5,17} Risk of stroke was found to be decreasing across the late menopausal age categories in trend-test ($P_{\text{trend}} = 0.0013$), though none of the estimates were found to be significant.⁵ Further, the decreasing risk across age at menopause categories was found for thrombotic stroke ($P_{\text{trend}} = 0.0007$) but not for haemorrhagic stroke ($P_{\text{trend}} = 0.3234$),⁵ suggesting inconsistency in finding across stroke subtypes. Stroke subtypes were inconsistently reported across medical insurance databases and hospital records, which could be the reason for the inconsistent findings.⁵

Association of reproductive lifespan, menarche and menopause with CVD and all-cause mortality

Reproductive lifespan

Overall, the association between reproductive lifespan and CVD mortality and all-cause mortality was conflicting. RLS was not significantly associated with CVD mortality,^{25–28} but it was significantly associated with 5% higher mortality for every additional year of RLS in one prospective study ($P = 0.03$).¹⁴ In this study, adjusting for BMI, waist circumference and other lifestyle factors only slightly attenuated the findings.¹⁴ In contrast, Yang et al. found that CVD mortality declined by 1% for every 1-year increase in RLS. Further, the risk of CVD mortality was lower among women in the normal weight and pre-obese categories compared with obese women¹⁷ (Table 2).

For stroke mortality, two large prospective studies from Japan showed a decreased risk for longer RLS,^{26,29} and adjusting for BMI and type of menopause only slightly attenuated the estimates.²⁶ However, two other prospective studies, which only included a low number of stroke cases, showed no association.^{25,28}

For all-cause mortality, the findings remained conflicting: decreased risk associated with longer RLS was found in one prospective study,³⁰ but an increased risk was found in another prospective study⁴ that treated RLS as a continuous exposure.¹⁴

Age at menarche adjusting for age at menopause

There was no clear evidence for the association of early menarche with CVD mortality,^{26,28,31,32} CHD mortality,^{26,28,31} stroke mortality^{26,28,31} and all-cause mortality^{28,30,32–34} (Table 2). A large prospective study of 423 460 women from Singapore showed the risk of CVD mortality was lower among non-smokers aged ≥ 17 years at menarche (hazard ratio [HR] 0.79, 95% CI 0.68–0.92) compared with those aged 13–14 years at menarche ($P_{\text{trend}} < 0.001$).³¹ The estimates remained changed after adjusting for menopausal status/age, oral contraceptives use and MHT use, BMI and other lifestyle factors.³¹ Additionally adjusting for diabetes and hypertension attenuated the estimates to HR 0.86

(95% CI 0.74–1.01),³¹ suggesting a modest effect of these co-morbidities on CVD mortality. The same study found consistent results for CHD mortality, but no association for stroke mortality. In contrast, Wu et al. found that early age at menarche was only associated with stroke mortality ($P_{\text{trend}} = 0.03$) but not associated with CHD mortality.²⁸

Further, three studies showed that women who experienced earlier age at menarche had a higher risk of all-cause mortality^{28,32,34} (Table 2).

Age at menopause adjusting for age at menarche

Overall, there was no sufficient evidence for an association between early menopause and risk of CVD mortality, CHD mortality and stroke mortality, although some evidence was found for the risk of all-cause mortality^{28,32,35–37} (Table 2). The risk of CVD mortality showed either null^{28,36,38} or positive association: risk of CVD mortality was 30% higher among women aged ≥ 55 years compared with those 50–54 years at menopause.³⁸ Risk of stroke mortality, however, showed null association in a large prospective study from the USA³⁷ and negative association (24% reduced risk) in those with later age at menopause (50–52 versus 49–50 years) in a larger prospective study from China.²⁸ Further, Hong et al. showed that risk of stroke mortality varied by the age at entry into the study. For example, the risk was higher among those aged ≥ 80 years at entry (HR 2.29, 95% CI 1.20–4.36) and remained null among those aged < 80 years at entry into the study (HR 0.61, 95% CI 0.18–2.11).³⁵ Such differences in the association by age could be due to the uneven distribution of causes of death among women. Particularly, cancer and CVD are the leading cause of death in younger and older women, respectively leading to higher incidence of CVD-related deaths in older compared with younger women.^{35,37}

Association of endogenous and total estrogen exposure with CVD events, and CVD and all-cause mortality

Endogenous estrogen exposure

There were only four studies taking pregnancy and use of OCP into account to estimate EEE^{9,14,15,39} (Table 3). One prospective study showed that longer duration of EEE was significantly associated with an increased risk of CVD mortality,¹⁴ whereas the other prospective study showed an association with a reduced risk of overall CVD mortality but no association with ischaemic heart disease or stroke mortality.⁹ No association was seen between EEE and all-cause mortality.^{14,15}

De Kleijn et al. found a lower risk among those who had > 18 years of EEE compared with ≤ 13 years for CVD mortality (HR 0.80, 95% CI 0.67–0.96), ischaemic heart disease

Table 2. Summary of studies examining associations between reproductive lifespan, age at menarche and age at menopause with CVD mortality and all-cause mortality

Outcomes	Short reproductive lifespan	Long reproductive lifespan	Early menarche	Late menarche	Early menopause	Late menopause
CVD mortality (<i>n</i> = 11 studies)						
Chang (2011) ²⁵	↔ (2–25 vs 26–32 years)	↔ (≥33 vs 26–32 years)	↔ (≤13 vs 14 years)	↔ (≥17 vs 14 years)	↑ (<40 vs 45–49 years)	↔ (≥50 vs 45–49 years)
Cui (2006) ²⁶	↔ (≤27 vs 31–33 years)	↔ (≥37 vs 31–33 years)				
Hong (2007) ³⁵	↔ (≤33 vs 36–37 years)	↔ (≥40 vs 36–37 years)				
Jansen (2002) ²⁷		↑ (per year)				
Jaspers (2017) ¹⁴						
Li (2013) ^{a36}						
Li (2013) ^{b36}						
Muller (2012) ^{b31}						
Tom (2012) ^{c38}						
Wu (2014) ²⁸	↔ (<27 vs 30–33 years)	↔ (≥35 vs <30–33 years)	↔ (<14 vs 15 years)	↔ (≥17 vs 15 years)	↔ (<40 vs 50–54 years)	↔ (≥55 vs 50–54 years)
Yang (2017) ¹⁷		↓ (per year)	↔ (≤11 vs 12–13 years)	↔ (≥14 vs 12–13 years)	↔ (<40 vs 50–54 years)	↔ (≥55 vs 50–54 years)
Zhang (2019) ³²						
CHD mortality (<i>n</i> = 7 studies)						
Chang (2011) ²⁵	↔ (2–25 vs 26–32 years)	↔ (≥33 vs 26–32 years)	↔ (≤13 vs 14 years)	↔ (≥17 vs 14 years)	↑ (<40 vs 45–49 years)	↓ (50–54 vs 45–49 years)*
Cui (2006) ²⁶	↔ (≤27 vs 31–33 years)	↔ (≥37 vs 31–33 years)			↔ (40–44 vs 45–49 years)	
Hong (2007) ³⁵						
Mondul (2005) ³⁷						
Muller (2012) ^{b31}						
Otuski (2018) ²⁹						
Wu (2014) ²⁸	↔ (<27 vs 30–33 years)	↓ (≥35 vs ≤28 years)*	↑ (≤12 vs 13–14 years)*	↓ (≥17 vs 13–14 years)*	↔ (<40 vs 45–49 years)	↔ (≥52 vs 49–50 yrs)
Stroke mortality (<i>n</i> = 7 studies)						
Chang (2011) ²⁵	↔ (2–25 vs 26–32 years)	↔ (≥33 vs 26–32 years)	↔ (<14 vs 15 years)	↔ (≥17 vs 15 years)	↔ (<40 vs 45–49 years)	
Cui (2006) ²⁶	↔ (≤27 vs 34–36 years)	↓ (≥37 vs 31–33 years)	↔ (≤13 vs 14 years)	↔ (≥17 vs 14 years)	↔ (<40 vs 45–49 years)	
Hong (2007) ³⁵						
Mondul (2005) ³⁷						
Muller (2012) ^{b31}						
Wu (2014) ²⁸	↔ (<27 vs <30–33 years)	↔ (≥35 vs <30–33 years)	↑ (<14 vs 15 years)*	↓ (≥17 vs 15 years)*	↔ (<40 vs 45–49 years)	↓ (50–52 vs 49–50 years)
Otuski (2018) ²⁹		↓ (<35 vs ≤28 years)*				

Table 2. (Continued)

Outcomes	Short reproductive lifespan	Long reproductive lifespan	Early menarche	Late menarche	Early menopause	Late menopause
All-cause mortality ($n = 11$ studies)						
Giles (2010) ³³	↔ (≤ 30 vs 31–39 years)	↔ (≥ 40 vs 31–39 years)	↔ (< 12 vs ≥ 12 years)		↑ (< 40 vs 45–49 years)	↔ (≥ 50 vs 45–49 years)
Hong (2007) ³⁵						
Jansen (2002) ²⁷	↔ (≤ 33 vs 36–37 years)	↔ (≥ 40 vs 36–37 years)			↑ (< 40 vs 50–54 years)	↓ (≥ 55 vs 50–54 years)*
Jaspers (2017) ¹⁴		↑ (per year)			↔ (< 40 vs 50–54 years)	↔ (≥ 55 vs 50–54 years)
Li (2013) ^{a36}					↑ (40–44 vs 45–49 years)*	↓ (50–54 vs 45–49 years)*
Li (2013) ^{b36}					↓ (< 40 vs 50–54 years)*	↓ (≥ 55 vs 50–54 years)*
Mondul (2005) ³⁷					↓ (< 45 vs 50–54 years)*	↑ (≥ 55 vs 50–54 years)*
Shadyab (2017) ^{d30}	↓ (< 33 vs 38–40 years)	↓ (> 40 vs 38–40 years)	↔ (< 12 vs ≥ 12 years)	↑ (18–20 vs 14 years)*	↑ (< 47 vs 49–50 years)	↓ (52 vs 49–50 years)*
Tamakoshi (2011) ³⁴			↑ (9–12 vs 14 years)		↑ (≤ 44 vs 45–54 years)	↓ (≥ 55 vs 45–54 years)*
Tom (2012) ^{c38}	↔ (< 27 vs 30–33 years)	↔ (≥ 35 vs < 30 –33 years)	↑ (< 14 vs 15 years)*	↓ (≥ 17 vs 15 years)*		
Wu (2014) ²⁸			↑ (≤ 11 vs 12–13 years)			
Zhang (2019) ³²						

↑ indicates the risk of outcome increased, ↓ indicates the risk of outcome decreased, ↔ indicates no evidence of association based on statistical significance at $P < 0.05$.

a: not receiving hormone replacement therapy, b: no smoker, c: natural menopause, d: all women.

*Trend of ordered exposure levels based on statistical significance at $P < 0.05$.

mortality (HR 0.84, 95% CI 0.64–1.10) and stroke mortality (HR 0.77, 95% CI 0.52–1.14) after adjusting for age. Although the effect sizes for CVD mortality were significant, ischaemic heart disease and stroke mortality yielded no statistical significance. Adjusting for socio-economic status, BMI and MHT marginally attenuated the association for all CVD subtypes.⁹ Similarly, in a large study of more than 300 000 participants across ten countries in Europe, the risk of all-cause mortality was lower in ≥ 34 years of EEE in the age-adjusted model (HR 0.89, 95% CI 0.82–0.97) compared with 0–22 years of EEE.¹⁵ However, the results were attenuated in the fully adjusted model for socio-economic and lifestyle factors (HR 0.98, 95% CI 0.91–1.06).¹⁵ With a limited number of studies, it remains inconclusive whether longer EEE was cardioprotective.

Total estrogen exposure

Studies further took postmenopausal MHT use into account to estimate TEE. Merz et al. used two methods to estimate TEE: adding time of postmenopausal MHT use but subtracting time of premenopausal OCP use and pregnancy from the reproductive lifespan; and adding time of MHT use, OCP use and pregnancy to the reproductive lifespan (Table 3).^{16,40} Under the first definition, there was no significant association with coronary artery disease.¹⁶ Under the second definition, adding OCP use and pregnancy revealed a decreasing age-adjusted prevalence and severity of coronary artery disease with increasing TEE. Further, the study showed that the association of TEE with age-adjusted prevalence and severity of coronary artery disease was mainly driven by MHT use.¹⁶

Adjusting for covariates

Most of the studies adjusted for age, education and lifestyle factors (including BMI and smoking). Six of 26 studies on estrogen exposure (RLS, EEE, TEE) adjusted for age at menopause/status,^{4,6,7,14,33,41} although only three studies adjusted for age at menarche^{4,28,30,40} (see Supplementary material, Table S6). The correlation coefficient of RLS with age at menopause ranged from 0.7 to 0.9, and with age at menarche was -0.03 to -0.20 .^{9,27} A higher percentage of studies adjusted for age at menopause (12 out of 34 studies) than age at menarche (17 out of 60 studies) (see Supplementary material, Tables S5–S7).

Meta-analysis

Seventeen cohort studies were included in the meta-analysis to compute pooled estimates, with hazard ratio from 14 studies, odds ratio from two studies, and risk ratio from one study. As the outcome measures were incident events, the pooled estimates were equivalent measures of relative risk (RR). Short reproductive lifespan was associated with increased risk of stroke events (RR 1.31, 95% CI 1.25–1.36;

Table 3. Summary of studies examining associations between EEE and TEE and CVD events, CVD mortality and all-cause mortality

Author (year)	Sample size	Formula of estrogen exposure	CVD	CVD mortality	All-cause mortality
Endogenous estrogen exposure (<i>n</i> = 4 studies)					
Farahmand (2019) ³⁹	3656	Age at menopause – age at menarche – pregnancy – OCP – breastfeeding – pregnancy-termination – postovulatory period ^a	↑ (T ₁ vs T ₃)		
Merritt (2015) ¹⁵	322 972	Age at recruitment/age at menopause – age at menarche – pregnancy – OCP ^b			↔ (≥34 vs 0–22 years) ↔ (per year)
Jaspers (2017) ¹⁴	1736	Age at menopause – age at menarche – pregnancy – OCP – breastfeeding – postovulatory period ^c		↑ (per year)	
De Kleijn (2002) ⁹	9450	Age at menopause – age at menarche – pregnancy – OCP – breastfeeding – miscarriage/stillborn – postovulatory period ^d		All CVDs ↓ (>18 vs ≥13 years) IHD ↔ (>18 vs ≥13 years) Stroke ↔ (>18 vs ≥13 years)	
Total estrogen exposure (<i>n</i> = 2 studies)					
Lee (2019) ⁴⁰	648	Age at menopause – age at menarche – OCP – pregnancy + HRT ^e	↔ (per year)		↔ (per year)
Lee (2019) ⁴⁰	648	Age at menopause – age at menarche + OCP + pregnancy + HRT ^f	↔ (per year)		↔ (per year)
Merz (2009) ¹⁶	646	Age at menopause – age at menarche – OCP – pregnancy + HRT ^e	70 carotid artery stenosis ↔ 50 carotid artery stenosis ↔ 70 carotid artery stenosis ↓ 50 carotid artery stenosis ↓		

Abbreviations: HRT, hormone replacement therapy; IHD, ischaemic heart disease; OCP, oral contraceptive pills; T, tertiles.

Notes ↑ indicates the risk of outcome increased significantly with estrogen exposure; ↓ indicates the risk of outcome significantly decreased with estrogen exposure; ↔ indicates no evidence of a significant association.

^aCalculated as the difference between the participant's age at menopause or age at CVD event or end of follow up – based on their first occurrence, and age at menarche. Further, pregnancy (40 weeks/per pregnancy), OCP, breastfeeding (duration of breastfeeding in months/per live birth), pregnancy termination (20 weeks/per termination), and postovulatory period (2 weeks/per cycle) were subtracted to yield EEE.

^bThe EEE was calculated as the difference between the participant's age at menopause (postmenopausal) and age at recruitment (either premenopausal, perimenopausal or of unknown menopausal status) and age at menarche, minus the length of time that a woman was pregnant or using OCPs.

^cThe total postovulatory period in months ((lifetime number of menstrual cycles* 2/4), subtracted from the cumulative EEE in months (age at menopause-age at menarche-pregnancy-breastfeeding-OCP), and finally expressed in years.

^dCalculated by subtracting age at menopause (in months) by age at menarche (in months). For each live birth, a further 9 months were subtracted. Further, for each miscarriage or stillborn child, 3 months were subtracted. And for each child, 4 months (the average duration of breastfeeding) were subtracted. Furthermore, the total duration of oral contraceptive use was subtracted. Finally, the total duration of the postovulatory period was subtracted to obtain the total duration of EEE.

^e[age at last cycle – age at menarche] – (premenopausal oral contraceptive time) – (pregnancy time) + (postmenopausal hormone therapy time), the study called it as a TEE.

^f[age at last cycle – age at menarche] + (premenopausal oral contraceptive use time) + (pregnancy time) + (postmenopausal hormone therapy time), the study called it as a supra-total estrogen exposure.

heterogeneity $I^2 = 0\%$, $P = 0.331$).^{5,6,42} There was no evidence for an association of short reproductive lifespan with CVD mortality (RR 1.02, 95% CI 0.89–1.15; heterogeneity $I^2 = 0\%$, $P = 0.543$)^{25,26,43,44} and all-cause mortality (RR 1.00, 95% CI 0.81–1.19; heterogeneity $I^2 = 75.3\%$, $P = 0.007$)^{27,28,30,33} (see Supplementary material, Figure S1). Similarly, early age at menarche (RR 1.17, 95% CI 0.20–2.14; heterogeneity $I^2 = 69.6\%$, $P = 0.070$)^{5,19} and early age at menopause (RR 1.13, 95% CI 0.71–1.56; heterogeneity $I^2 = 22.9\%$, $P = 0.255$)^{5,19} were not significantly associated with moderately higher risk of stroke events. There was no evidence that early age at menarche was associated with CVD mortality (RR 1.05, 95% CI 0.95–1.14; heterogeneity $I^2 = 0.2\%$, $P = 0.391$).^{26,28,31,32} For all-cause mortality, the association with age at menarche (RR 1.08, 95% CI 0.97–1.19; heterogeneity $I^2 = 77.4\%$, $P = 0.001$)^{26–28,31} and early age at menopause (RR 1.05, 95% CI 0.95–1.15; heterogeneity $I^2 = 79.8\%$, $P = 0.000$)^{28,30,32,35–38} did not reach statistical significance. Early age at menopause also had no evidence for an association with CVD mortality (RR 1.08, 95% CI 0.98–1.18; heterogeneity $I^2 = 0\%$, $P = 0.431$)^{28,32,35,36,38} (see Supplementary material, Figure S1).

Discussion

Main findings

This study found that a longer RLS was associated with a reduced risk of CVD events particularly stroke. Meta-analyses of prospective studies further revealed that women with short reproductive lifespan had 31% increased risk of stroke (RR 1.31, 95% CI 1.25–1.36). Further, the role of EEE and TEE was unclear: decreasing risk of all-cause mortality⁹ and severity of CAD¹⁶ and increasing risk of all-cause mortality.¹⁴

Strengths and limitations

This is the first comprehensive systematic review and meta-analysis to explore women's reproductive lifespan and risk of CVD events and mortality, including the mutual effect of age at menarche and age at menopause – i.e. the effect of age at menarche adjusted for age at menopause. A small number of studies included in the meta-analysis may limit the interpretation of the pooled relative risks for reproductive lifespan, age at menarche and age at menopause. However, we have described individual effect estimates in details in the results section for better understanding of true associations. There could be a chance of reporting bias, as only studies with positive results could have been published.

Interpretation

Most studies defined estrogen exposure as the interval between the timing of menarche and menopause. Also a few

studies further accounted for the duration of breastfeeding, pregnancy, stillbirths/miscarriages, OCP and MHT, yielding three indices of estrogen exposure.⁴⁵ It was not revealed, however, which of these indices was the optimal measure of estrogen exposure. A previous study of 3476 Dutch participants revealed that addition of extra variables such as duration of breastfeeding and duration of use of OCP did not improve the predictive accuracy of the model assessing the relationship between EEE (calculated as the interval between age at menarche and menopause) and bone mineral disease disorders.⁴⁵ Calculating length of estrogen exposure is dependent on several factors, such as regularity in menstrual cycles, length of non-ovulatory cycling, number of pregnancies, number of miscarriages and breastfeeding duration, which are prone to recall bias.¹⁶

Imprecision in measurement of these factors will lead to measurement bias.⁹ Therefore, robust epidemiological studies with measurement of estrogen exposure and associated health risk would strengthen the evidence.

The underlying mechanism linking a longer reproductive lifespan with reduced risk of CVD events can be attributed to prolonged exposure to endogenous estrogen. Estrogen affects the blood vascular system that keeps blood vessels flexible and regulates the level of metabolic markers such as lipids, inflammatory markers and coagulant system.⁴⁶ Previous epidemiological studies have revealed that longer reproductive lifespan is associated with reduced risk of hip fracture mortality,⁴⁷ bone mineral diseases,^{47,48} type 2 diabetes⁴⁹ and Parkinson's disease.⁵⁰ Further, longer endogenous estrogen exposure is associated with reduced risk of cardiovascular mortality.⁹ So far, no prospective studies have examined the relationship with estrogen exposure indices in CVD context.

Conclusion

Reproductive lifespan was the most commonly used method for estrogen exposure measurement. We found evidence of increased risk of CVD events, particularly stroke for those with a shorter reproductive lifespan. Association for EEE and TEE remained unclear due to limited availability of studies.

Conflicts of interests

None. Completed disclosure of interests forms are available to view online as supporting information.

Authors' roles

SRM conducted the literature review, and analysed and drafted the article. SRM and HFC performed the searches and screening of articles. SRM, HFC, MW and GDM contributed to the conceptualisation of study. All authors contributed to the interpretation of results and critical revision

of the article. All the authors approved the final version of article.

Funding

The study received no external funding. GDM is supported by the Australian National Health and Medical Research Council Principal Research Fellowship (APP1121844).

Acknowledgement

None.

Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Figure S1. Forest plots for showing meta-analyses of pooled estimates for the association between (A) short reproductive lifespan, (B) early age at menarche and (C) early age at menopause and the risk of stroke events, cardiovascular disease mortality and all-cause mortality.

Table S1. Quality assessment based on the Newcastle–Ottawa scale.

Table S2. Characteristics of the studies assessing the relationship of estrogen exposure, age at menarche and age at menopause with risk of cardiovascular disease events and mortality.

Table S3. Association between age at menarche and risk of cardiovascular disease events and mortality (studies not adjusting for age at menopause/status).

Table S4. Association between age at menopause/menopausal status and risk of cardiovascular disease events and mortality (studies not adjusting for age at menarche).

Table S5. Adjusted variables for studies assessing the relationship of estrogen exposure with risk of cardiovascular disease events and mortality.

Table S6. Adjusted variables for studies assessing the relationship of age at menarche with risk of cardiovascular disease events and mortality.

Table S7. Adjusted variables for studies assessing the relationship of age at menopause/menopausal status with risk of cardiovascular disease events and mortality.

Appendix S1. Search strategy for PubMed, EMBASE and Web of Science.

Appendix S2. Supplementary methods. ■

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